MBUNeXt: Multibranch Encoder Aggregation Network Based on Layer-Fusion Strategy for Multimodal Brain Tumor Segmentation

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***Abstract*—Multimodal brain tumor segmentation (BraTS), integrated with surgical robots and navigation systems, enables accurate surgical interventions while maximizing the preservation of surrounding healthy brain tissue. However, multimodal brain scans suffer from large interclass differences in brain tumor subregions and information redundancy, leading to inadequate fusion of multimodal information and significantly affecting the accuracy of BraTS. To address the above problems, we propose a multibranch encoder aggregation (MEA) network based on a layer-fusion strategy called multibranch UNeXt (MBUNeXt). The network comprises three well-designed modules: the multimodal feature attention (MFA) module, the MEA module, and the large-kernel convolution skip (LCS)-connection module. These modules work together to achieve precise segmentation of brain tumors. Specifically, the MFA module preserves the intermodality similarity structure through attention mechanisms and Gaussian modulation functions, thereby filtering redundant information. Then, the MEA module exploits the correlations among multiple modalities to effectively integrate multimodal hybrid feature representation and optimize multimodal information fusion. In addition, the LCS module constructs multiple groups of depthwise separable convolutions with large kernel, which can guide the network to attend to features at different scales, thereby addressing the issue of significant interclass differences in brain tumor subregions. The experimental results on the large-scale public datasets, BraTS2019 and BraTS2021, which consist of approximately 5000 3-D brain scans, demonstrate that our proposed method has achieved SOTA performance, with average Dice scores of 85.84% and 91.11%, respectively. It also performs well on the BraTS-Africa2024 dataset with low imaging quality, confirming its robustness. The code is available at https://github.com/liuqinghao2018/MBUNeXt**

***Index Terms*—Brain tumor segmentation (BraTS), layer-fusion strategy, magnetic resonance imaging (MRI), multimodal fusion.**

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I. Introduction

BRAIN tumors can cause severe neurological defects and even pose a life-threatening risk. In modern biomedical image processing, segmentation [1], [2], [3] techniques are widely utilized for the diagnosis, staging, and follow-up of brain tumors. Magnetic resonance imaging (MRI) [4] is a widely adopted imaging modality in the automated diagnosis of brain tumor diseases due to its ability to provide high-contrast images of soft tissues. With different parameter acquisitions, a range of MRI sequences with different modalities can be obtained [5]. For the diagnosis of brain tumor diseases [6], MRI modalities commonly utilized include T1-weighted (T1), post-contrast T1-weighted (T1Gd), T2-weighted (T2), and T2 fluid-attenuated inversion recovery (FLAIR). Among them, T1 can reveal various tomographic anatomical structures; T1Gd can visualize areas with enhanced blood supply after the injection of a contrast agent, providing further insight into the tumor's internal characteristics; T2 is related to the water content in tissues and organs, and can precisely locate the area of the lesion; FLAIR can present the surrounding conditions of the tumor site, showing the edema area. Different imaging modalities can present complementary information regarding the lesion area and surrounding tissues. Therefore, the use of multimodal medical images can make up for the shortcomings of single-modality images, effectively helping doctors to accurately identify the surgical area and lesion structure. The integration of medical image analysis techniques with the application of surgical robot systems in modern medicine enables more precise and refined surgical interventions, maximizing the preservation of surrounding normal brain tissue and enhancing the safety and efficacy of the surgery [7], [8], [9].

However, the vast amount of medical data associated with multimodal MRI scans has greatly increased the workload of disease diagnosis. Doctors need to invest additional time in manually assessing an extensive volume of medical images, which can result in diagnostic errors. Hence, there is an urgent need to develop a multimodal MRI segmentation method that can effectively leverage the complementary information present in multimodal images and extract regions of interest (lesion tissues and organs) from these images.

In recent years, significant progress has been made in the field of deep learning, with various deep neural networks demonstrating versatility and efficiency in medical image segmentation. These networks have outperformed traditional algorithms, yielding notable results. U-Net [10], a symmetric encoder–decoder network structure, has been

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particularly influential in medical image segmentation. Its encoder–decoder architecture preserves image details, contributing to improved segmentation accuracy, and has become a leading network in the field. Khan et al. [11] integrated deep saliency maps and an improved dragonfly algorithm, enhancing the accuracy of brain tumor detection and classification through feature enhancement, selection, and extreme learning machine (ELM) classification. Rajinikanth et al. [12] employed VGG-UNet-assisted segmentation and support vector machine (SVM) classification to enhance the detection and classification accuracy of brain tumors in MRI slices through feature extraction, selection, and fusion. Maqsood et al. [13] used a deep neural network and multiclass SVM (M-SVM), enhancing the precision of brain tumor detection and classification through image contrast enhancement, customized convolutional neural networks (CNNs) segmentation, transfer learning-based feature extraction, and entropy-based feature selection. Badjie and Ülker [14], based on the deep transfer learning architecture of AlexNet, improve the accuracy of brain tumor detection and classification in MR images through feature enhancement, selection, and an optimized network structure. In addition, TransBTS [15] employs 3-D CNNs to extract spatial features and uses a Transformer for global feature modeling, ultimately achieving image segmentation through a convolutional decoder. This method differs from traditional approaches that rely only on CNNs, as it combines the strengths of local feature extraction with global information. It demonstrates excellent performance in tasks such as multimodal brain tumor segmentation (BraTS) and classification.

Taking into account the characteristics of multimodal MRI scans, and the advantages and disadvantages of existing methods, we design a multibranch encoder aggregation (MEA) network based on a layer-fusion strategy to achieve accurate segmentation of brain tumor. Specifically, three well-designed modules are introduced into the U-shaped network to solve practical problems in multimodal BraTS, such as inadequate multimodal information fusion, information redundancy, and large interclass differences of brain tumor subregions.

To effectively integrate the multimodal image information obtained from the multibranch encoder, we propose a multimodal feature attention (MFA) module, which utilizes an attention mechanism to guide the fusion process and assist the multibranch encoder to extract multimodal information. Importantly, by embedding Gaussian functions in the MFA, the module cannot only emphasize mutual features of multimodal images, but also filter out subrelated information.

To aggregate the unique features of different modal images, an MEA module is introduced, which connects the multibranch encoder and decoder. The module converts the multimodal features extracted from the multibranch encoder into tokens and inputs them into the Transformer for global multimodal information modeling.

To guide the network to focus on the target area, effectively identify lesions of varying sizes, and address the issue of size variation, a large-kernel convolution skip (LCS)-connection module is designed. In the LCS module, multiple sets of depthwise separable convolutions with large kernel are constructed to achieve a larger receptive field and capture multiscale context information [16], [17].

Integrating these modules into the network architecture, we propose a new architecture called multibranch UNeXt (MBUNeXt).

The experimental results of MBUNeXt on several highly challenging multimodal BraTS datasets, particularly for BraTS2019 and BraTS2021, provide compelling evidence of the effectiveness of our model, which surpasses other existing BraTS methods and demonstrates its outstanding performance in this field. To further validate its robustness, we have conducted experiments on the BraTS-Africa2024 dataset as well, achieving good segmentation results and proving the method's strong robustness.

In summary, the contributions of this work are as follows.

1) A U-shaped network named MBUNeXt is proposed for multimodal BraTS. The network is designed to emphasize and aggregate the potential shared features of multimodal images and filter out redundant information, while achieving a larger receptive field at a global scale, enabling the network to capture multiscale feature information, emphasizing target areas, and accurately locating lesions.
2) The MFA module is proposed to emphasize potential shared features and filter out redundant information, which integrates multimodal feature maps in the channel dimension and constructs Gaussian modulation functions to adaptively adjust the activation response of the segmentation target. The proposed MEA module fully aggregates multimodal information and learn complementary features between multiple modalities by modeling nonlinear relationships within and between modalities. The LCS module addresses the issue of significant interclass scale differences between brain tumor subregions. It achieves this by employing multiple sets of depthwise separable convolutions with a large kernel, allowing for the extraction of lesion feature information at different scales.
3) Extensive experiments were conducted on two large-scale multimodal MRI BraTS tasks (BraTS2019 and BraTS2021), which included approximately 5000 3-D brain scans. To further validate the robustness of the method, experiments were also carried out on the BraTS-Africa2024 dataset. The results demonstrate that the proposed method achieved superior outcomes compared to the current SOTA approaches, even with fewer parameters, proving its robustness and broad applicability.

II. Related Work

A. Medical Image Segmentation Methods

Utilizing the strong feature extraction capability of CNNs, the fully convolutional network (FCN) [18] employs CNNs for end-to-end semantic segmentation. U-Net [10] has been widely used for a variety of medical image segmentation tasks and has inspired a great deal of research to adapt to specific segmentation needs or improve performance. Subsequently, many networks were designed that use this U-shaped architecture as the backbone, such as Res-UNet [19], UNet++ [20], and KiU-Net [21], which further advanced the progress of medical image segmentation. nnU-Net [22] demonstrated the effectiveness of a generalized U-Net architecture with minimal modifications in achieving remarkable results across

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various vision tasks, adapting and processing diverse datasets automatically. The extending nn-UNet [23] framework by incorporating group normalization and axial attention mechanisms, achieving precise BraTS through adaptive architecture optimization in the BraTS 2021 challenge and securing first place in the competition. FANet [2] leverages feedback attention to iteratively optimize prediction masks and incorporates a run-length encoding strategy. Experimental results on multiple public datasets demonstrate its superior performance compared to existing methods. However, CNN-based network models often face challenges in integrating global information and handling multiscale features.

Recent Transformer-based models excel in computer vision by modeling long-range dependencies. In medical image segmentation, U-shaped architectures integrated with Transformers [24] achieve SOTA performance. For example, UNETR [25] designed a U-shaped network that combines a Transformer and CNN. It utilizes ViT [26] as an encoding module to extract contextual information and employs CNN to capture local information, leading to the final segmentation results. Swin UNETR [27], on the other hand, uses Swin Transformer [28] as an encoder to compute self-attention in a shifted window and also enables further processing of the extracted multiscale features by patch merging. Swin UNETR has demonstrated exceptional performance in a wide range of medical image segmentation tasks. DeepCRC-SL [1] constructs a 1-D colorectal coordinate system to transform labels into coordinate maps. It enhances the segmentation continuity through auxiliary regression tasks, differentiates tissues using self-attention layers, and adopts a coordinate-driven self-learning strategy to integrate unlabeled data for performance improvement. It performs remarkably in colorectal cancer and colorectum segmentation. HResFormer [3] combines 2-D and 3-D Transformers, utilizes the hybrid local–global fusion module (HLGM) to integrate information, and adopts the residual learning of the hybrid model. It outperforms existing methods in multiple medical image segmentation benchmark tests. However, these methods [15], [29], [30], [31] incorporate self-attention mechanisms into the encoding process, which demands larger datasets to effectively learn input sequence relationships for improved output prediction.

B. Multimodal Fusion Methods

The purpose of multimodal fusion is to utilize the information from various modalities of medical images to assist in data representation, enhance the discriminative capability of networks, and thereby improve the accuracy of multimodal medical image segmentation. Based on deep learning, fusion strategies [32] can be categorized into three types: input-level fusion, layer-level fusion, and decision-level fusion.

In the input-level fusion strategy, these methods fuse multimodal images channel by channel into multichannel inputs, learn the feature representation after fusion, and then train the segmentation network. Three-dimensional ShiftBTS [33] extends ShiftViT to 3-D multimodal medical images, introducing a 3-D shift operation, enhanced feature extractor (EFE), and cascaded attention module. It achieves superior performance in complex BraTS while maintaining computational efficiency. Zhou et al. [34] proposed a multitask segmentation network on the BraTS dataset, which fused multimodal MRI scans channel by channel, learned the feature representation after fusion in the input space, decomposed the BraTS task into three different but related subtasks, and gradually trained them. This method achieved the best performance on the BraTS2015 and BraTS2017 datasets. Most of these methods focus on subsequent complex segmentation network architecture design, without considering the relationships between different modalities, and without analyzing how to fuse different feature information to improve segmentation performance.

In the layer-level fusion strategy, these methods utilize two or more modal images as a single input to train segmentation networks. Subsequently, the learned feature representations from these inputs are fused across the layers of the network, and the fused results are then fed back to the decision layer to obtain the final segmentation output. Dolz et al. [35] proposed a 3-D fully CNN based on DenseNets, which extended dense connections to accommodate multimodal segmentation. Each imaging modality has a pathway containing layers with dense connections, enabling the network to learn more complex feature representations. It demonstrated SOTA performance in multimodal brain tissue segmentation challenges. M^2FTrans [36] framework integrates a modality-masked fusion Transformer with learnable tokens and masked self-attention mechanisms, optimizing multimodal feature fusion through spatial-weighted attention and channel-wise fusion Transformers. It achieves significant performance improvements over existing methods on the BraTS dataset, particularly excelling in critical scenarios where dominant modalities are absent. AAHN [37] proposes an asymmetric dual-stream encoder with Transformer-CNN branches, employs the Transformer-CNN feature alignment and fusion (T-CFAF) module and cross-modality heterogeneous graph fusion (CMHGF) module for adaptive feature fusion, and achieves superior performance on six medical image segmentation datasets. PDMINet [38] introduces a prototype-driven framework with multimodal information interaction, prototype-guided fusion, and multi-expert integration to address feature aliasing in BraTS. It achieves superior performance on BraTS datasets by enhancing subregion-specific features and suppressing irrelevant information. However, the large number of dense connections within each pathway introduces a significant number of parameters, making the network difficult to train. Moreover, while multimodal medical images provide information at different levels, they also introduce redundant data and substantial noise, which may degrade network performance when using overly simplistic dense connections.

Decision-level fusion focuses more on fusion issues because each modality is treated as an input to a separate network. These networks [39], [40] can learn the complex and complementary feature information of each modality, and finally, fusion is performed at the output layer of the network to obtain the segmentation results. Nie et al. [41] proposed a novel multi-FCNs network architecture dedicated to infant brain tissue segmentation, employing a multimodal network training and high-level feature fusion strategy to achieve more accurate segmentation. In general, compared to input-level fusion and layer-level fusion, if the fusion method is effective enough, decision-level fusion can achieve better segmentation performance. However, decision-level fusion requires multiple network architectures to individually extract information from each modality, leading to an exponential parameter increase. This demand for more memory and computational resources limits its practical application. Hence, adopting an appropriate

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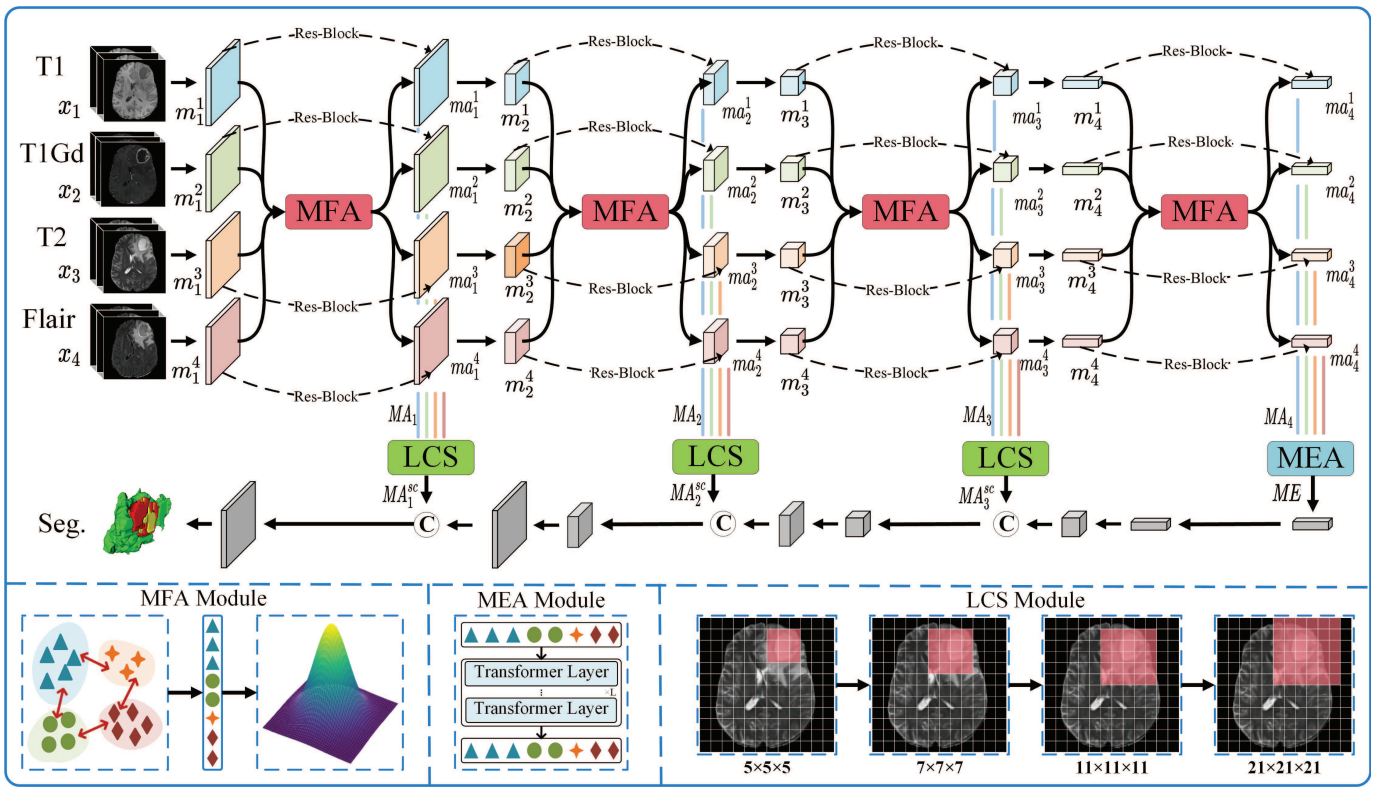


Fig. 1. Framework of the proposed MBUNeXt for BraTS. Specifically, MBUNeXt is mainly composed of four parts: the back-bone structure of MBUNeXt, the MFA module, the MEA module, and the LCS module.

fusion strategy is crucial as it not only fully exploits the complementary advantages of multimodal medical images but also effectively eliminates redundant information and noise interference, significantly enhancing the effectiveness and value of medical image fusion.

III. Method

A. Overview

MBUNeXt is a novel U-shaped network that incorporates multiple encoding branches and one decoding branch. The overall architecture of the MBUNeXt network is illustrated in Fig. 1. Specifically, MBUNeXt consists of four main components: the multibranch encoder, the MFA module, the MEA module, and the LCS module. This section provides an overview of the general structure of MBUNeXt and further elaborates on significant implementation details of the model, such as how the MFA module effectively emphasizes potential shared features, how the MEA module fully aggregates multimodal complementary information, and how the LCS module guides the network to focus on feature information at different scales.

B. Multibranch Encoder

Unlike most variants of U-Net, the MBUNeXt network employs a layer-fusion strategy to divide the encoding path into multiple branches, with each branch processing a single modality MRI input x_j ($j \in \{1,2,3,4\}$) through an identical network structure. Each encoder branch consists of four stages $i \in \{1,2,3,4\}$, where each stage applies a convolutional operation $\mathrm{En_i}$ (comprising a convolutional layer, InstanceNorm, and PReLU activation) followed by downsampling via a $2 \times 2 \times 2$ kernel with stride 2 to halve the feature resolution, avoiding pooling. Residual connections are introduced to mitigate gradient dissipation during backpropagation. Given multimodal MRI inputs $x_j \in \mathbb{R}^{1\times H\times W\times D}$, the network progressively encodes them into deep features m_i^j. The output of MFA is multiplied by the original feature m_i^j and then element-wise added to m_i^j, generating the enhanced feature ma_i^j and forming a residual-like mechanism. The relationships are mathematically defined as

$$m_i^j = \begin{cases} \mathrm{En}_1\left(x_j\right), & i = 1 \\ \mathrm{En}_i\left(\mathrm{ma}_{i-1}^j\right), & i = 2,3,4 \end{cases} \tag{1}$$

$$\mathrm{ma}_i^j = m_i^j + \mathrm{MFA}\left(m_i^j\right) \times m_i^j \tag{2}$$

$$M_i = \mathrm{concat}\left(m_i^1, m_i^2, m_i^3, m_i^4\right) \tag{3}$$

$$\mathrm{MA}_i = \mathrm{concat}\left(\mathrm{ma}_i^1, \mathrm{ma}_i^2, \mathrm{ma}_i^3, \mathrm{ma}_i^4\right) \tag{4}$$

where $m_i^j, \mathrm{ma}_i^j \in \mathbb{R}^{k\times(H/2^i)\times(W/2^i)\times(D/2^i)}$. MFA denotes the multimodal feature attention module, M_i represents the multimodal features, $\mathrm{MA_i}$ denotes the enhanced multimodal features processed by MFA, and $M_i, \mathrm{MA}_i \in \mathbb{R}^{4k\times(H/2^i)\times(W/2^i)\times(D/2^i)}$. Here, k indicates the number of feature channels.

C. Multimodal Feature Attention

The multibranch encoder consists of mutually independent encoding paths that extract unique sensitive features and a substantial amount of information from each modality. However,

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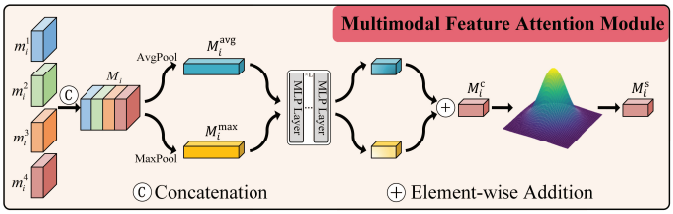


Fig. 2. MFA module uses channel attention and Gaussian distribution function to emphasize potential shared features and filter out redundant information.

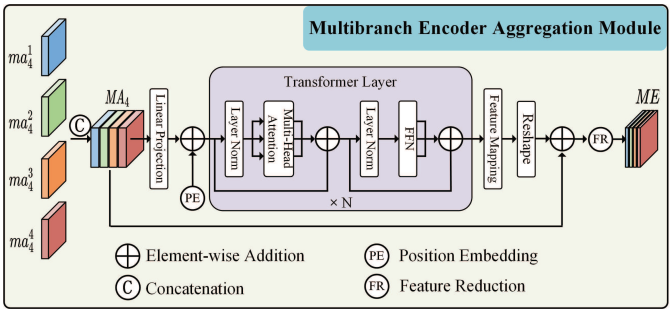


Fig. 3. MEA module aggregates multiple branches of the encoder by Transformer to achieve global context modeling and fusion of multimodal information.

not all of this information contributes to the segmentation of lesions; instead, it introduces redundant information. Moreover, in the process of capturing unique features associated with each modality, the multibranch encoders tend to overlook those multimodal potential shared features that are also important for the segmentation regions. Therefore, the MFA module is designed to emphasize the shared information among different modalities and filter out redundant information. Its purpose is to assist the multibranch encoder in extracting crucial regions for image segmentation. Fig. 2 illustrates the structure of the MFA module.

Specifically, we focus on a specific part given an input image by exploiting the feature relationships between channels. In the calculation of channel attention [42], [43], [44], we consider that maximum pooling collects the most prominent and salient features from different modalities, and mean pooling collects shared features from different modalities. To aggregate the channel information of the feature map, we employ both mean pooling and maximum pooling operations. This results in the generation of two sets of features: mean pooled channel features denoted as $M_i^{\text{avg}} \in R^{4k\times1\times1\times1}$ and maximum pooled channel features denoted as $M_i^{\max} \in R^{4k\times1\times1\times1}$, as shown in the following equation:

$$M_i^{\text{avg}} = \text{AvgPool}\left(M_i\right) \tag{5}$$

$$M_i^{\max} = \text{MaxPool}\left(M_i\right). \tag{6}$$

Then, M_i^{avg} and $M_i^{\max}$ are passed through a compression network consisting of a multilayer perceptron (MLP). In order to fully integrate multimodal information, the enhancement ratio is set e, and the channel attention map is first expanded to $M_i^{c\prime} \in \mathbb{R}^{4ek\times1\times1\times1}$, and then gradually compressed to generate a multimodal channel attention map $M_i^c \in \mathbb{R}^{k\times1\times1\times1}$.

The enhancement ratio e is an adjustable parameter and e is set to [16, 8, 4, 2] in the MFA module for the different encoding layers. Subsequently, we merge the average-pooling channel features M_i^{avg} and the max-pooling channel features $M_i^{\max}$ using element-wise summation.

In this module, it is noteworthy that the Gaussian distribution function [45], [46], [47] is utilized as a modulation function to reactivate the distribution of features by learning the attention weights of each channel, which helps to model the correlation between different channels explicitly and adaptively adjust the activation response for the segmentation target. The implementation process can be expressed as

$$M_i^s = \mathcal{G}\left(M_i^c\right) \tag{7}$$

where M_i^c denotes the multimodal channel attention map, M_i^s denotes the multimodal attention scale, and $\mathcal{G}$ represents the Gaussian distribution function. $\mathcal{G}$ maps the Gaussian distribution by calculating the mean and variance of all activation values in M_c, which can be expressed as

$$\mu = \frac{1}{k}\sum_{n=1}^{k}\left(M_i^c\right), \sigma = \sqrt{\frac{1}{k}\sum\nolimits_{n=1}^{k}\left(M_i^c-\mu\right)^2} \tag{8}$$

where μ represents the mean of the attention map, and σ represents the standard deviation of the attention map. The output of the MFA module is $M_i^s \in \mathbb{R}^{k\times1\times1\times1}$.The overall MFA module can be summarized as

$$M_i^s = \mathcal{G}\left(\text{MLP}\left(M_i^{\text{avg}}\right) + \text{MLP}\left(M_i^{\max}\right)\right). \tag{9}$$

This method optimizes the multimodal attention map by using the modulation function, which helps the attention blocks to focus on the common areas in the feature map, emphasizing the multimodal potential shared information while suppressing useless information and noise.

D. Multibranch Encoder Aggregation

Different encoding branches are employed to process distinct MRI modalities separately. The multibranch encoder extracts the unique information of each modality step by step to obtain deep feature maps of multiple modalities. In most methods, multiple feature maps are fused by concatenation, addition, and multiplication.

Such fusion methods are both prone to information loss and difficult to fully aggregate multimodal information. Given the capability of self-attention mechanisms to capture long-range dependencies and contextual information [26], we incorporate the Transformer as the MEA module, as depicted in Fig. 3. Specifically, the MEA module captures the multimodal feature map MA_4 from the multibranch encoder.

First, the MEA module flattens the feature map of each modality from the spatial dimension and depth dimension to one dimension and combines them to obtain a token sequence with dimension $4k \times N$. In order to achieve precise localization of the lesion and accurate segmentation, learnable positional encoding is employed, which can be expressed by the following equations:

$$z_0 = \text{LP}\left(\text{MA}_4\right) + \text{PE} \tag{10}$$

where LP denotes the linear projection layer, PE $\in \mathbb{R}^{4k\times N}$ denotes the location embedding layer, and $z_0 \in \mathbb{R}^{4k\times N}$ denotes the feature token sequence, k refers to the number of feature channels for each modality; and N denotes the length of the

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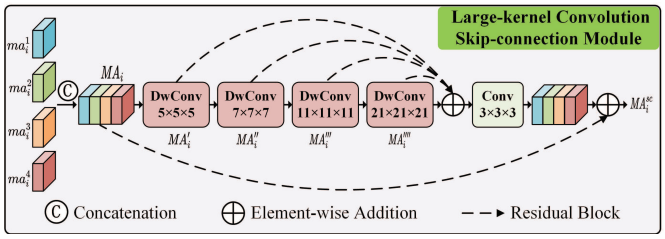


Fig. 4. LCS module adopts a set of large-size depthwise separable convolutions to expand the perceptual field and enhance edge features.

token sequence after flattening the feature maps from the spatial and depth dimensions into a 1-D sequence.

The MEA module utilizes an L-layer Transformer architecture, where each Transformer layer is composed of multihead attention and an MLP. Within the lth layer of the MEA module, the outputs are computed as follows:

$$z_l' = \mathrm{MSA}\left(\mathrm{LN}\left(z_{l-1}\right)\right) + z_{l-1} \tag{11}$$

$$z_l = \mathrm{MLP}\left(\mathrm{LN}\left(z_l'\right)\right) + z_l' \tag{12}$$

where LN represents layer normalization, MSA represents the multihead self-attention layer, and z_l represents the output of the lth layer Transformer.

The key component of the Transformer architecture is the self-attention mechanism, which leverages the relationships between Q, K, and V to capture dense correlations between tokens and capture global information. The calculation process of self-attention is as follows:

$$\mathrm{Attention}\left(Q, K, V\right) = \mathrm{Softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right)V \tag{13}$$

where Q, K, and V denote queries, keys, and values, respectively; d represents the size of the query and key.

In the MEA module, the self-attention mechanism models nonlinear relationships within and between modalities, fully integrating multimodal information and learning complementary relationships between all modalities. After processing through the self-attention mechanism, the features undergo convolutional operations to reduce the number of channels, followed by residual connections to obtain the final multimodal fusion feature ME

$$\mathrm{ME} = \mathrm{FR}\left(\mathrm{Reshape}\left(z_L\right) + \mathrm{MA}_4\right) \tag{14}$$

where $\mathrm{ME} \in \mathbb{R}^{k\times(H/8)\times(W/8)\times(D/8)}$, FR represents a feature reduction block composed of convolutional operations, which integrates channel compression ($4k \to k$), and Reshape reconstructs spatial dimensions from flattened token sequences.

E. Skip-Connection With Large-Kernel Convolution

In the context of medical image segmentation, the primary objective of neural networks is to accurately identify and segment the lesion area. However, in brain tumors, there are often several different tissue subregions with large differences in size and shape. Accurately identifying the location and boundaries of morphologically diverse lesions poses a challenge for networks utilizing a single conventional $3 \times 3 \times 3$ convolution. To overcome these difficulties, we design the LCS module, as shown in Fig. 4. By incorporating a series of large-scale depthwise separable convolutions, the LCS module expands the receptive field, allowing the network to capture the intricate structures of lesions spanning various sizes. To further compress network parameters [48], we utilize $1 \times 1 \times n$, $1 \times n \times 1$ and $n \times 1 \times 1$ depthwise strip convolutions to approximately replace the standard $n \times n \times n$ depthwise convolution.

Specifically, the LCS module extracts feature maps from the multibranch encoder. First, $\mathrm{MA_i}$ ($i \in \{1, 2, 3\}$) pass through depthwise separable convolutions of $5 \times 5 \times 5$, $7 \times 7 \times 7$, $11 \times 11 \times 11$, and $21 \times 21 \times 21$ in sequence, as shown in the following equation:

$$\mathrm{MA}_i' = \mathrm{Conv}^{dw}_{5\times5\times5}\left(\mathrm{MA}_i\right) \tag{15}$$

$$\mathrm{MA}_i'' = \mathrm{Conv}^{dw}_{7\times7\times7}\left(\mathrm{MA}_i'\right) \tag{16}$$

$$\mathrm{MA}_i''' = \mathrm{Conv}^{dw}_{11\times11\times11}\left(\mathrm{MA}_i''\right) \tag{17}$$

$$\mathrm{MA}_i'''' = \mathrm{Conv}^{dw}_{21\times21\times21}\left(\mathrm{MA}_i'''\right) \tag{18}$$

where $\mathrm{MA}_i', \mathrm{MA}_i'', \mathrm{MA}_i''', \mathrm{MA}_i'''' \in \mathbb{R}^{4k\times(H/2^i)\times(W/2^i)\times(D/2^i)}$ represent the outputs of the depthwise separable convolution, and $\mathrm{Conv}^{dw}_{n\times n\times n}$ represents a depthwise separable convolutional layer with kernel size $n \times n \times n$. Spatial information is extracted by these multiscale convolutions. In the depthwise separable convolution, the spatial feature relationship and interchannel feature relationship of the multimodal image are computed completely independently. Therefore, a series of depthwise separable convolution blocks of different sizes extract feature maps for each modality separately. Finally, this information is summed up and fed into a regular convolution block for multimodal information integration, as expressed by the following equations:

$$\mathrm{MA}_i^{\mathrm{sc}} = \mathrm{Conv}_{3\times3\times3}\left(\mathrm{MA}_i + \mathrm{MA}_i' + \mathrm{MA}_i'' + \mathrm{MA}_i''' + \mathrm{MA}_i''''\right) \tag{19}$$

where $\mathrm{MA}_i^{\mathrm{sc}} \in \mathbb{R}^{2k\times(H/2^i)\times(W/2^i)\times(D/2^i)}$ represents the output of the LCS module and is fed into the decoder at each decoding stage, $\mathrm{Conv}_{3\times3\times3}$ represents a regular convolutional layer with kernel size $3 \times 3 \times 3$.

Through the LCS module, the shallow features from corresponding positions in the encoder and the deep features from the decoder are effectively aggregated during the up-sampling process at each level of the network. This allows the decoder to leverage more high-resolution information, facilitating a more accurate recovery of fine details in the original image. More importantly, the LCS module provides multiscale and multilevel multimodal information for image segmentation to enhance edge features, and thus obtains finer segmentation results.

F. Decoder

The decoder obtains the multimodal feature map ME that has been meticulously modified by the MEA module, gradually restores it to the original 3-D image size ($H \times W \times D$), and produces segmentation results. Specifically, at each stage, the encoder features from the LCS module are concatenated with those corresponding to the decoder. The concatenated features are processed by $3 \times 3 \times 3$ convolutions, and then upsampled by transposed convolutions until they are restored to the original input resolution. Finally, the features at the original image resolution are processed through a sigmoid activation function and a $1 \times 1 \times 1$ convolutional layer for generating a multimodal medical image segmentation map.

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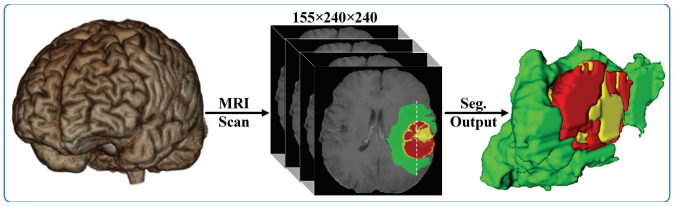


Fig. 5. In the utilized BraTS datasets, annotations include different brain tumor subregions: the NCR (red), the ED (green), and the ET (yellow).

IV. Experiment

The proposed MBUNeXt model, along with other SOTA segmentation methods, underwent training and testing on two challenging multimodal brain tumor datasets, namely, the BraTS2019 and BraTS2021 datasets. In addition, the BraTS-Africa2024 dataset was incorporated into the validation process, allowing for the assessment of the MBUNeXt model's robustness across datasets with unique imaging characteristics using identical experimental settings. The segmentation methods were evaluated based on common performance metrics. Furthermore, the effectiveness of the modules proposed within MBUNeXt was demonstrated through ablation studies.

A. Dataset and Implementation

1) BraTS2019 Dataset: This dataset [49], [50], [51], provided by the BraTS 2019 Challenge, focuses on glioma segmentation and includes multimodal MRI data from 19 institutions. The training set consists of 335 cases, each with 3-D MRI scans in four modalities: T1, T1Gd, T2, and T2-FLAIR. Images are resized to 155 × 240 × 240 voxels. Labels include the nonenhancing tumor core (NCR), peritumoral edema (ED), and enhancing tumor (ET), forming three evaluation regions: the whole tumor (WT, including NCR, ED, and ET), the tumor core (TC, including NCR and ET), and ET, as shown in Fig. 5. We adopted Swin UNETR [27]'s data partitioning method, splitting the training set into an 80%:20% ratio and training networks using fivefold cross-validation. Our method was compared with various SOTA approaches for a comprehensive evaluation.

2) BraTS2021 Dataset: This dataset [50], [51], [52] was provided by the BraTS 2021 challenge and included additional routine clinically acquired 3-D multimodal MRI scans. It consists of a total of 5004 3-D brain scans. Apart from the number of volumes, all other information regarding the BraTS2019 and BraTS2021 datasets remained identical. In the BraTS2021 challenge, the training set consisted of 1251 cases. Compared to the training set of BraTS2019, BraTS2021 contained approximately four times more data. This significant increase in data volume was a crucial factor in selecting these two datasets for our experiments. It allows us to demonstrate that our network could still achieve high performance despite the substantial difference in the amount of available data.

3) BraTS-Africa2024 Dataset: This dataset [53] is derived from the MICCAI-CAMERA-Lacrina Fund BraTS-Africa 2024 Challenge, with a training set consisting of 60 cases. Similar to the BraTS2019 dataset, each case includes MRI scans of four modalities and shares the same segmentation targets. However, unlike BraTS2019, this dataset is characterized by the widespread use of low-quality MRI technology, resulting in poorer image quality and resolution, which poses additional challenges for the segmentation task.

4) Experiment Details: The MBUNeXt model was implemented based on PyTorch and trained with an NVIDIA Quadro RTX A6000 GPU. During the training phase, a batch size of 2 was set, and the model was trained for 300 epochs. The Dice loss function was employed to update the network weights. The AdamW optimizer was utilized with an initial learning rate of 0.0001, and it underwent linear warm-up and cosine annealing to gradually adjust the learning rate. In addition, various general data augmentation techniques were applied. Specifically, during the training process, when trained with the BraTS2019, BraTS2021, and BraTS-Africa2024 datasets, the input images were randomly cropped to 128 × 128 × 128 voxels. For the 3-D images, random mirroring flips were performed along the axial, coronal, and sagittal planes with a probability of 0.5. In addition, random intensity shift within the range of (−0.1, 0.1) and random intensity scale within the range of (0.9, 1.1) were applied to the images. Prior to training, all images were normalized to have zero mean and unit standard deviation. The Dice loss function was employed to update the network weights, which is defined as

$$L_{\text{Dice}} = 1 - \frac{2\,|S \cap G|}{|S| + |G|} \tag{20}$$

where S denotes the predicted segmentation result and G denotes the ground truth.

5) Performance Measurement: We employed the same evaluation metrics as other SOTA networks, such as Swin UNETR [27] and TransBTS [15], which include the average Dice coefficient (Dice) and 95% Hausdorff Distance (HD95). The detailed definition of the average Dice coefficient is as follows:

$$\text{Dice} = \frac{2\,|S \cap G|}{|S| + |G|} \tag{21}$$

where S denotes the predicted segmentation result and G denotes the ground truth. A larger Dice score represents better segmentation performance.

The detailed definition of the Hausdorff Distance (HD) metrics is as follows:

$$\text{HD}\left(G', P'\right) = \max\left\{\max_{g' \in G'} \min_{p' \in P'} \left\|g' - p'\right\|, \max_{p' \in P'} \min_{g' \in G'} \left\|p' - g'\right\|\right\}. \tag{22}$$

Among them, G' represents the ground truth surface point set; and P' represents the predicted surface point set. g' represents a point that belongs to the set G'; p' represents a point that belongs to the set P'. The HD95 is measured based on the 95th percentile of distances between the ground truth point set and the predicted point set. This approach reduces the impact of outliers during the calculation of HD.

B. Experiments on BraTS2019

In the BraTS2019 experiments, we conducted fivefold cross-validation to compare MBUNeXt with various existing advanced algorithms, including U-Net [10], nnU-Net [22], nnFormer [30], TransBTS [15], UNETR [25], and Swin UNETR [27]. The average evaluation results of all the methods in fivefold cross-validation are presented in Table I. In

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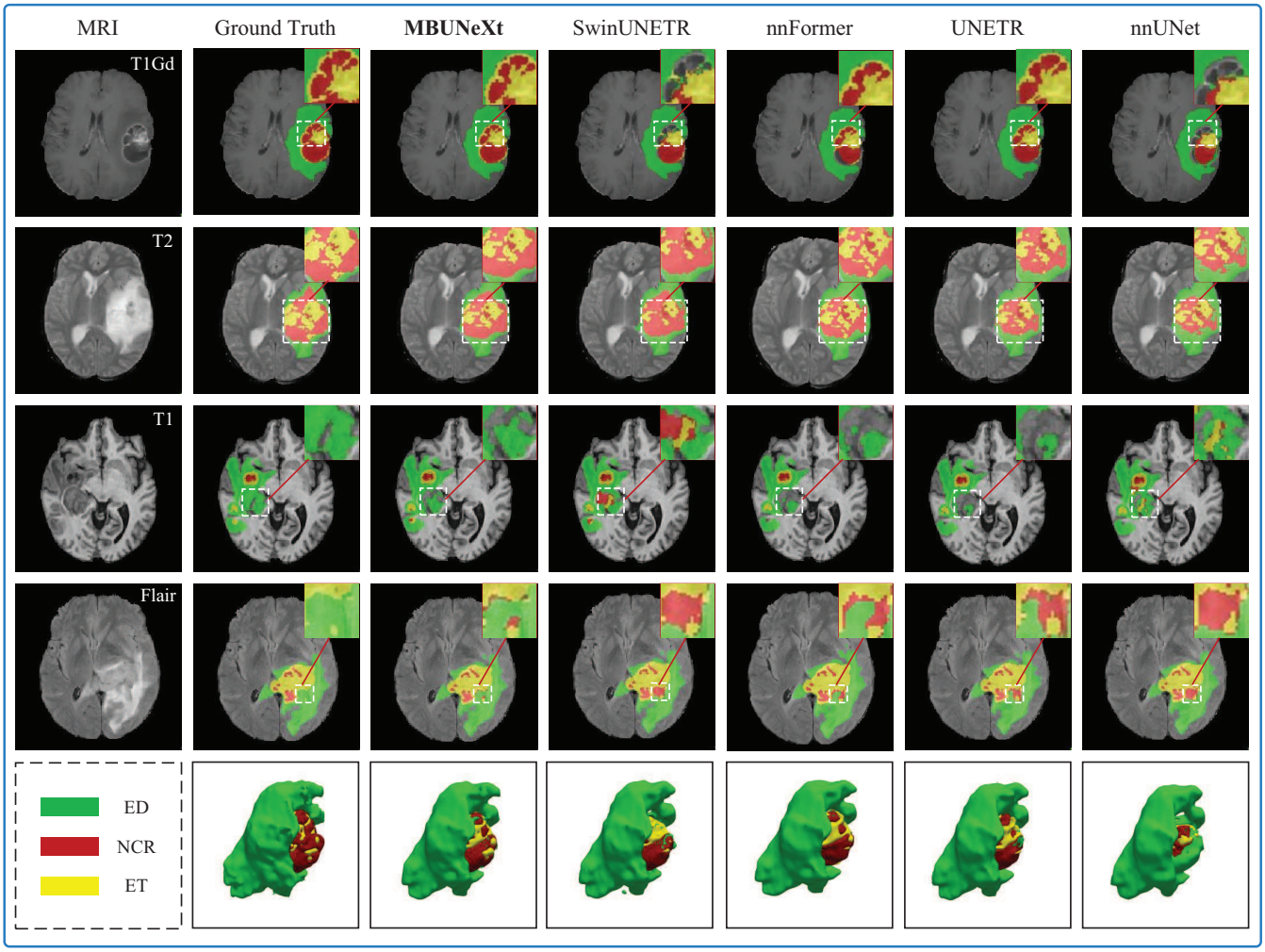


Fig. 6. Visual comparison of BraTS results in BraTS2019 and BraTS2021. For the purpose of visualization, we selected certain modalities that clearly depict the lesions.

experiments, MBUNeXt demonstrated the best performance in terms of average Dice coefficients for all subregions of brain tumors, outperforming other CNN-based and Transformer-based baselines. Specifically, MBUNeXt achieved Dice scores of 85.42%, 90.62%, and 81.48% for the TC, WT, and ET regions, respectively, and also had the smallest average HD95 for the three segmentation target regions, achieving the SOTA performance. The segmentation outputs of several networks are illustrated in Fig. 6. MBUNeXt demonstrated excellent performance in BraTS. The remarkable performance of the proposed method can be attributed to the effective extraction of unique features from each modality by the multibranch encoder, the assistance of the MFA module in capturing multimodal information, the learning of complementary relationships between all modalities through the self-attention mechanism in the MEA module, and the provision of multiscale and multilevel multimodal information for multimodal BraTS by the LCS module, resulting in a more refined segmentation effect.

In general, the experimental results showed that MBUNeXt outperformed other SOTA methods, such as Swin UNETR and nnFormer, in terms of Dice scores and average HD95 for the ET, WT, and TC regions. This demonstrates the effectiveness of the proposed module in MBUNeXt and its ability to leverage the rich information from multimodal brain tumor datasets for improved segmentation performance.

C. Experiments on BraTS2021

In comparison to the BraTS2019 dataset, the dataset used in the BraTS2021 Challenge was updated to include a larger number of clinical MRI scans. Specifically, the BraTS2021 dataset contained approximately four times the amount of data compared to the training set of BraTS2019. In the BraTS2021 experiments, we conducted fivefold cross-validation to compare MBUNeXt with several SOTA methods. The average evaluation results of all the methods in the fivefold cross-validation are presented in Table I. We trained the network model using the hyperparameters defined in the BraTS2019 experiments. The MBUNeXt achieved average Dice scores of 91.52%, 93.18%, and 88.62% for the TC, WT, and ET regions, respectively, and also had the smallest average HD95 for the three segmentation target regions, achieving the SOTA performance. These results also demonstrate the robust performance of our network with an increased amount of data.

In addition, it is observed that Transformer-based networks have competitive benchmarks on the BraTS2021 dataset. The

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TABLE I

AVERAGE RESULTS OF SOTA METHODS ON THE BRATS2019 AND BRATS2021 TRAINING DATASETS USING FIVEFOLD CROSS-VALIDATION. THE BOLDED RESULTS INDICATE THE BEST SCORES FOR EACH TUMOR SUBREGION

Method	Brats2019								Brats2021							
	Dice Score(%)				HD95				Dice Score(%)				HD95			
	TC	**WT**	**ET**	**Avg.**	**TC**	**WT**	**ET**	**Avg.**	**TC**	**WT**	**ET**	**Avg.**	**TC**	**WT**	**ET**	**Avg.**
U-Net	78.09	80.98	72.27	77.11	12.71	18.12	9.59	13.47	83.53	87.49	81.21	84.08	10.95	14.71	7.39	11.01
TransBTS	79.72	87.39	76.72	81.28	12.20	16.67	8.77	12.54	88.00	90.70	85.31	88.00	**8.30**	10.76	6.37	8.47
nnU-Net	80.58	88.26	76.77	82.06	12.42	**15.36**	10.18	12.65	88.37	91.43	86.19	88.66	9.87	13.68	7.56	10.37
UNETR	80.95	89.24	77.87	82.69	17.66	31.01	17.51	22.06	89.05	91.83	86.58	89.15	12.13	17.57	10.48	13.40
nnFormer	84.37	87.15	79.09	83.54	15.31	26.54	14.90	18.92	89.60	90.95	87.54	89.36	11.12	18.44	7.37	12.31
Swin UNETR	85.25	90.50	80.95	85.57	13.39	23.92	13.63	16.98	91.46	93.08	88.44	90.99	9.25	16.50	6.86	10.87
MBUNeXt	**85.42**	**90.62**	**81.48**	**85.84**	**11.81**	16.65	**6.74**	**11.74**	**91.52**	**93.18**	**88.62**	**91.11**	8.37	**10.60**	**6.20**	**8.39**

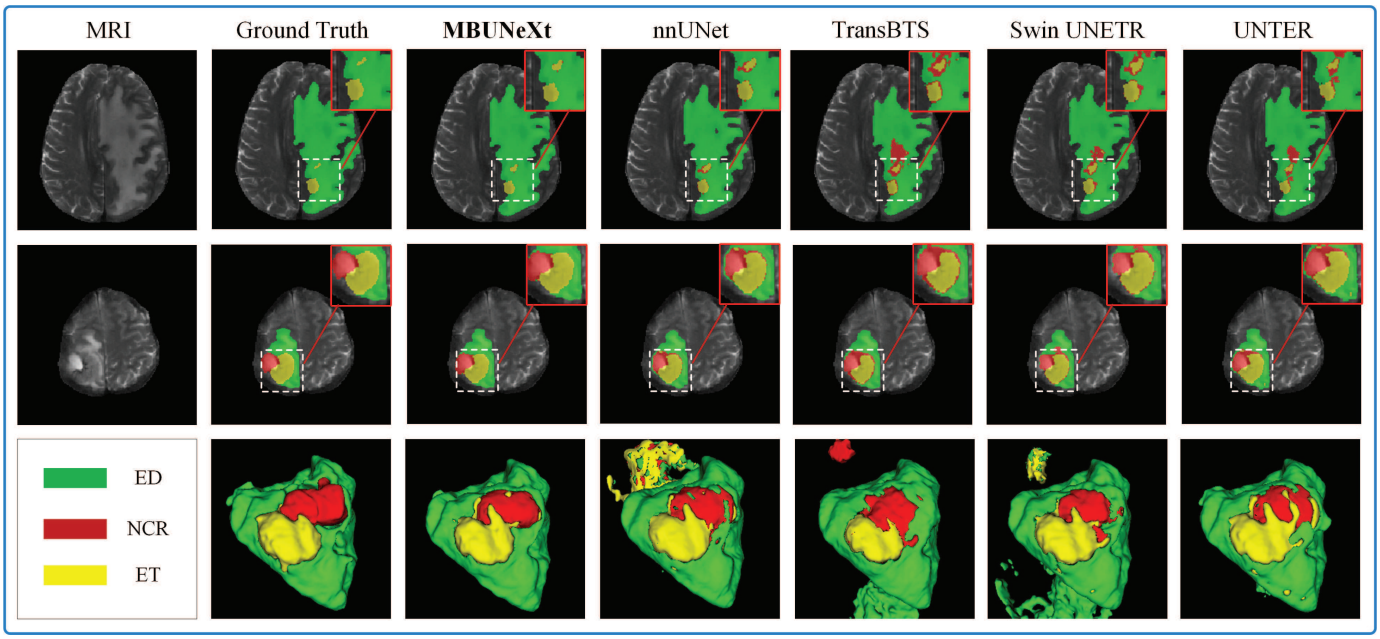


Fig. 7. Visual comparison of segmentation results in the BraTS-Africa2024 dataset. MBUNeXt significantly reduces mis-segmentation, demonstrating its superior performance on low-quality imaging data.

TABLE II

FIVEFOLD CROSS-VALIDATION AVERAGE RESULTS OF SOTA METHODS ON THE BRATS-AFRICA2024 DATASET. BOLD RESULTS INDICATES THE BEST SCORE FOR EACH SUBREGION

Method	Dice Score(%)				HD95			
	TC	**WT**	**ET**	**Avg.**	**TC**	**WT**	**ET**	**Avg.**
U-Net	73.44	69.51	70.32	71.09	23.69	52.57	21.03	32.43
nnFormer	71.56	83.04	67.72	74.11	27.71	23.84	29.13	26.89
UNETR	73.70	87.25	76.05	79.00	25.01	22.06	19.51	22.20
Swin UNETR	73.99	86.99	77.27	79.42	30.14	19.02	22.19	23.78
TransBTS	76.82	86.34	76.38	79.85	27.53	19.77	23.08	23.46
nnU-Net	77.51	87.53	78.43	81.16	23.31	18.32	18.29	19.97
MBUNeXt	**80.29**	**89.23**	**79.57**	**83.03**	**20.91**	**15.38**	**17.96**	**18.08**

limitations observed in the Transformer structure could be attributed to its lack of certain intrinsic inductive biases that are present in CNNs, such as translational invariance and local correlation. Therefore, the Transformer-based networks require a larger volume of data to effectively learn the relationships within the input sequences and make accurate predictions. Consequently, Transformer-based models tend to exhibit better performance on large-scale datasets. In contrast, MBUNeXt achieves excellent segmentation performance by extracting unique multimodal feature representations through CNN multibranch encoders and utilizing multiple designed modules to assist in extracting shared features.

D. Experiments on BraTS-Africa2024 Dataset

To evaluate the robustness of the MBUNeXt model, validation was conducted on the BraTS-Africa2024 dataset, which is crucial for establishing the model's generalization capability

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TABLE III

COMPARISON OF NUMBER OF PARAMETERS, FLOPS, AND INFERENCE TIME FOR VARIOUS MODELS IN BRATS2019 EXPERIMENTS

Models	Params(M)	FLOPs(G)	Inference Time (s)
TransBTS	32.99	264.28	2.25
nnU-Net	19.71	437.63	1.81
UNETR	102.45	203.24	1.68
nnFormer	149.68	771.45	1.70
Swin UNETR	62.19	790.59	5.71
MBUNeXt	32.02	712.66	5.28

*Parameters and FLOPs are calculated on 128×128×128 inputs. Inference time is averaged over sliding-window inference on 155×240×240 images from BraTS2019.

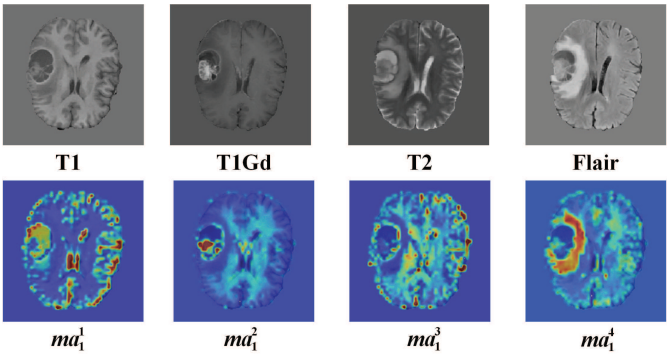


Fig. 8. Visualization of feature maps from each encoder in the MBUNeXt.

under varying medical imaging quality conditions. As shown in Table II, the results on the BraTS-Africa2024 dataset demonstrate that the MBUNeXt model not only performs excellently on datasets with high imaging quality (such as BraTS2019 and BraTS2021) but also maintains strong competitiveness on the lower quality BraTS-Africa2024 dataset. Fig. 7 illustrates the segmentation results of multiple networks, showing that MBUNeXt effectively reduces mis-segmentation. MBUNeXt achieved average Dice scores for the TC, WT, and ET regions on the BraTS-Africa2024 dataset that were 2.78%, 1.70%, and 1.14% higher than the suboptimal method, respectively, and also attained the smallest average HD95 values across all three segmentation target regions, demonstrating SOTA performance. This further confirms the versatility and broad applicability of the model in the field of multimodal segmentation.

E. Computational Efficiency

In Table III, we present the number of parameters, FLOPs, and inference time of various models in the BraTS2019 benchmark. MBUNeXt has 32.02M parameters, 712.66G FLOPs, and an average inference time of 5.28 s. Compared to Transformer-based models (Swin UNETR, nnFormer, and UNETR), MBUNeXt demonstrates significantly fewer parameters—a 78.6% reduction compared to nnFormer and a 48.5% reduction relative to Swin UNETR—while maintaining competitive performance. Although its FLOPs are higher than those of UNETR and TransBTS, MBUNeXt achieves a better balance between model complexity and performance compared to CNN-based models (e.g., nnU-Net). These results highlight the potential of MBUNeXt to deliver robust performance with reduced model size, though at the cost of increased computational load compared to lightweight architectures.

F. Ablation Studies

Next, we performed ablation experiments on the BraTS2019 training dataset to assess the influence of fusion strategies and individual modules on the overall performance of MBUNeXt. The Dice coefficient was adopted as the primary evaluation metric for this purpose.

The backbone network, designed in conjunction with a layer-fusion strategy, plays a crucial role in achieving SOTA performance. Table IV presents the performance differences between networks employing the input-level fusion strategy, the layer-level fusion strategy, and the decision-level fusion

TABLE IV

QUALITATIVE COMPARISON OF DIFFERENT FUSION STRATEGIES ON THE BRATS2019 TRAINING DATASET

Architecture	Dice Score(%)			
	TC	**WT**	**ET**	**Avg.**
Single branch encoder (input-level fusion)	76.01	82.25	75.35	77.87
Multibranch encoder (layer-level fusion)	**82.81**	**89.57**	**80.22**	**84.20**
Multiple encoder-decoder (decision-level fusion)	77.45	87.28	77.85	80.86

strategy. All networks utilized the same encoder and decoder architecture, and shared identical parameter settings. Through comparison, it can be observed that the average Dice score of the multibranch encoder network was 6.33% higher than that of the single-branch encoder network, indicating a significant improvement in segmentation performance. In addition, by visualizing the features of each encoder in the multibranch network (see Fig. 8), it is evident that each individual encoder can better extract unique features of the corresponding modality. In addition, as shown in Table V, among the four modalities (T1, T1Gd, T2, and Flair), "Missing 1 Modality" represents the average Dice scores across the four scenarios where any single modality is missing (e.g., T1 absent and T1Gd absent), resulting in an average Dice of 65.14% for MBUNeXt versus 63.36% for Swin UNETR. "Missing 2 Modalities" corresponds to the average Dice results when any two modalities are missing (six combinations in total), where MBUNeXt achieves 41.14% compared to Swin UNETR's 40.94%. For "Missing 3 Modalities" (four combinations), MBUNeXt demonstrates superior robustness with an average Dice of 20.04%, significantly outperforming Swin UNETR's 15.30%. Experimental results demonstrate that MBUNeXt consistently outperforms Swin UNETR across all modality missing scenarios, validating its modality robustness. Notably, the performance gap widens as more modalities are missing-for instance, MBUNeXt achieves a 4.74% absolute gain over Swin UNETR under three-modality absence. This improvement is attributed to the ability of each encoder branch in the multibranch network to extract modality-specific features and capture complementary aspects of the input data. Therefore, we adopted the U-shaped network with a multibranch encoder as the backbone network.

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TABLE V

PERFORMANCE COMPARISON BETWEEN MBUNEXT AND SWIN UNETR UNDER MISSING MODALITIES

Input	Method	Dice Score(%)			
		TC	**WT**	**ET**	**Avg.**
Missing 1 Modality	Swin UNETR	61.20	**79.67**	49.20	63.36
	MBUNeXt	**65.23**	78.79	**51.40**	**65.14**
Missing 2 Modalities	Swin UNETR	**39.60**	58.66	24.56	40.94
	MBUNeXt	38.71	**59.21**	**25.49**	**41.14**
Missing 3 Modalities	Swin UNETR	16.24	24.81	4.85	15.30
	MBUNeXt	**17.11**	**36.33**	**6.69**	**20.04**

TABLE VI

ABLATION STUDY OF THE MFA MODULE

Architecture	Dice Score(%)			
	TC	**WT**	**ET**	**Avg.**
Backbone	82.81	89.57	80.22	84.20
Backbone + MFA (without Gaussian distribution)	83.87	90.19	80.10	84.72
Backbone + MFA (with Gaussian distribution)	**84.23**	**90.27**	**80.70**	**85.07**

TABLE VII

ABLATION STUDY OF DIFFERENT MULTIBRANCH ENCODER AGGREGATE METHODS IN THE MEA MODULE

Aggregate method	Dice Score(%)			
	TC	**WT**	**ET**	**Avg.**
Multiplication	71.60	82.81	70.31	74.91
Summation	78.00	86.24	77.75	80.66
Concatenation	82.81	89.57	80.22	84.20
MEA	**83.67**	**90.39**	**80.25**	**84.77**

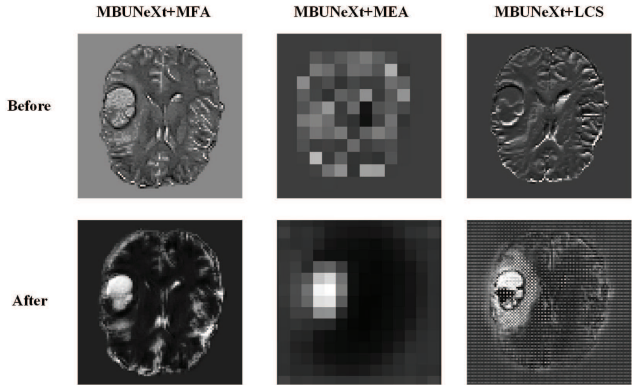


Fig. 9. Visualization comparison of feature maps before and after processing by various modules.

TABLE VIII

ABLATION STUDY OF MODULES

Models	Dice Score(%)			
	TC	**WT**	**ET**	**Avg.**
Baseline	82.81	89.57	80.22	84.20
Baseline+MFA	84.23	90.27	80.70	85.07
Baseline+MEA	82.62	89.14	81.01	84.26
Baseline+LCS	83.14	89.76	80.42	84.44
Baseline+MFA+MEA	84.72	90.59	81.37	85.56
Baseline+MFA+LCS	84.55	90.17	80.98	85.23
Baseline+MEA+LCS	83.93	90.38	81.31	85.21
Baseline+MFA+MEA+LCS	**85.42**	**90.62**	**81.48**	**85.84**

Table VI presents the performance improvements achieved by incorporating the designed MFA module into the backbone network, as well as the impact of using the Gaussian distribution function as the modulation function in the MFA module. Compared to the backbone network of MBUNeXt, the introduction of the MFA module without the Gaussian distribution resulted in an average Dice score improvement of 0.52% for multimodal BraTS. Furthermore, it can be observed that when the MFA module with the Gaussian distribution was introduced, an improvement of 0.87% was achieved in BraTS compared to the backbone network of MBUNeXt. This demonstrates the ability of the MFA module to emphasize potential shared features while filtering out redundant information.

Table VII compares different multibranch encoder aggregate methods, including concatenation, summation, multiplication, and the proposed MEA module. The MEA module creatively utilized Transformer as an MEA module to leverage its powerful capability in modeling multimodal nonlinear relationship and capturing global contextual information. As presented in Table VII, our approach achieved the best results in each segmentation category, surpassing the commonly used concatenation method by 0.57% (the backbone network employed the concatenation method). This indicates that the MEA module effectively integrated multimodal information and learned the complementary relationships between modalities.

Table VIII illustrates the performance evolution as MFA, MEA, and LCS modules are incrementally integrated. The results indicate that the MFA module significantly improves segmentation accuracy in the TC and WT regions, while the MEA module excels in enhancing boundary delineation for the ET region. When combined, the MFA achieves shared feature reinforcement and redundancy suppression through channel-wise feature fusion and Gaussian modulation, while the MEA explores complementary features via intramodal/intermodal nonlinear modeling, forming a progressive feature optimization workflow, outperforming other dual-module combinations in terms of performance. As shown in Fig. 9, further incorporation of the LCS module expands the receptive field via large-sized convolution kernels ($5\times5\times5$, $7\times7\times7$, $11\times11\times11$, and $21 \times 21 \times 21$), thereby optimizing the segmentation performance in the TC region and enhancing the extraction capability of multiscale lesion features in the ET region. The hierarchical integration of the three-level modules validates the effectiveness of the proposed MBUNeXt architecture through a multimodal encoding framework in achieving precise tumor subregion segmentation.

G. Limitation and Future Work

The core objective of this study is to address the challenge of multimodal BraTS, and thus, our proposed method relies on aligned multimodal images from the same patient as network inputs. However, this requirement limits the direct application of the method to datasets containing only single-modality images. To address this limitation, future work will

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focus on designing a multimodal framework compatible with missing modalities, which dynamically models intermodal feature interactions to directly accommodate single-modality or partially incomplete inputs, thereby enhancing the method's generalizability and robustness.

V. Conclusion

This article presents a novel MEA network based on a layer-fusion strategy for multimodal BraTS. The network incorporates three well-designed modules to enhance its segmentation performance. The proposed MFA module fuses the multimodal features of multibranch encoders at different stages, emphasizes the potential shared information in different modalities, and filters out redundant information. And the MEA module aggregates multibranch encoders, model the nonlinear relationship among different modalities, and effectively integrating multimodal information. In addition, the LCS module guides the network in attending to feature information at various scales, enabling accurate identification of lesions of different sizes and addressing the issue of size variability. Comprehensive experiments were conducted on the BraTS2019, BraTS2021, and BraTS-Africa2024 datasets to evaluate the effectiveness of the proposed method. The experimental results show that MBUNeXt achieves SOTA performance in multimodal BraTS with a smaller number of parameters, by effectively extracting features from each modality and comprehensively fusing multimodal information, while also demonstrating good robustness.

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